

End-sem exam

● Graded

Student

Samyak Sanghvi

Total Points

21.75 / 30 pts

Question 1

(no title)

3.25 / 5 pts

1.1 — 1a

1 / 1 pt

✓ + 0.5 pts Nothing happens

✓ + 0.5 pts because histogram is already uniform after first equalization

+ 0 pts No/Wrong answer

1.2 — 1b

0.5 / 1 pt

✓ + 0.25 pts Y corresponds to luminance

+ 0.25 pts as Y is weighted sum of R, G, B

✓ + 0.25 pts C_B and C_R corresponds to chrominance

+ 0.25 pts as they are differences in R, G, B

+ 0 pts No/Wrong answer

1.3 — 1c

1 / 1 pt

✓ + 0.5 pts Fourier transform $T = B^2$

✓ + 0.5 pts $T = \text{sinc}^2$

+ 0 pts No/Wrong answer

1.4 — 1d

0.25 / 1 pt

+ 1 pt Closing with small SE

+ 1 pt Thresholded median filter

+ 0 pts No/Wrong answer

💬 + 0.25 pts But it will modify other pixels as well which are not pepper noise

1.5 — 1e

0.5 / 1 pt

+ 0.25 pts In Freq dom, after Gaussian: $H_g.H.F + H_g.N$
Give only when 2 is absent

+ 0.5 pts After inverse filtering: $H_g.F + H_g.N/H$

+ 0.25 pts $H_g.N = 0$ (high freq noise multiplied with low pass gaussian is 0 in entire frequency)
Give only when 4 is absent

✓ + 0.5 pts Conclude: Original recovered image is Gaussian blurred

+ 0 pts No/Wrong answer

Question 2

(no title)

5 / 5 pts

2.1 — 2a

2 / 2 pts

✓ + 0.5 pts s_0 is constant

✓ + 0.5 pts $s_0 = 1/\sqrt{8}$

✓ + 0.5 pts s_1 is half-period of cosine = $0.5 \cdot \cos(\pi(n+0.5)/8)$

✓ + 0.5 pts s_1 decreases from $1/2$ to $-1/2$

– 0.5 pts If not(x-axis should be integers in $[0, 7]$)

+ 0 pts No/Wrong answer

2.2 — 2b

2 / 2 pts

✓ + 0.5 pts $\phi_{11} = \sqrt{2}$ in $[1/2, 1]$

✓ + 0.25 pts $\phi_{11} = 0$ elsewhere

✓ + 0.5 pts $\chi_{20} = 2$ from $[0, 1/8)$

✓ + 0.5 pts $\chi_{20} = -2$ from $[1/8, 1/4)$

✓ + 0.25 pts $\chi_{20} = 0$ elsewhere

+ 0 pts No/Wrong answer

2.3 — 2c

1 / 1 pt

✓ + 0.5 pts DCT better for smooth signal because mostly low frequencies

✓ + 0.5 pts Haar wavelets better for piecewise continuous signal because nonzero coeffs only at discontinuities

– 0.5 pts No explanation

+ 0 pts Incorrect or no answer

Question 3

(no title)

2.5 / 5 pts

3.1 — **3a**

2 / 2 pts

✓ + 0.5 pts Step 1 split [0, 0.2, 1]

✓ + 0.5 pts Step 2 split [0, 0.04, 0.2]

✓ + 0.5 pts Step 3 split [0.04, 0.072, 0.2]

✓ + 0.5 pts Decoded message: abb

+ 0 pts No/Wrong answer

3.2 — **3b**

0.5 / 3 pts

✓ + 1 pt H channel has 0°/360° discontinuity at red

✓ - 0.5 pts Explanation for H channel is missing or wrong

+ 1 pt Block DCT will have large high-freq coeffs

- 0.5 pts Only mentioned discontinuity / loss of spatial redundancy, not high frequencies in DCT

+ 1 pt Coeffs will not be quantized to zero / RLE will not be effective

+ 0 pts Incorrect or no answer

Question 4

(no title)

4.5 / 5 pts

4.1 — 4a

2 / 2 pts

✓ + 0.5 pts Dilation twice = dilation with 5x5 square SE

✓ + 0.5 pts Justify: max over adjacent of adjacent pixels

✓ + 0.5 pts Closing twice = closing once with the same SE

✓ + 0.5 pts Justify: Closing is idempotent

+ 0 pts No/Wrong answer

+ 0.25 pts If SE is bigger than 3
Give only when 1 is absent

4.2 — 4b

2 / 2 pts

✓ + 1 pt Opening with disk of appropriate size

✓ + 1 pt Difference between opening with $r = 20$ and opening with $r = 26$

– 0.5 pts Erosion or closing instead of opening

+ 1 pt Partially correct: Hit-or-miss transform (won't detect disks of radius 20 near another disk)

+ 0 pts Incorrect

4.3 — 4c

0.5 / 1 pt

+ 1 pt Peaks of distance transform / medial axis transform

+ 1 pt Other valid solution e.g. isolated pixels after erosion

+ 0.5 pts Hough transform (may find false positives as well)

✓ + 0.5 pts Other partially correct solution

+ 0 pts Incorrect or no answer

temp2 will be empty because temp1 contains only disks of size $< b_i$

Question 5

(no title)

3 / 5 pts

5.1

5a

2 / 3 pts

+ 1 pt 2D Hough space (cx, cy) or (r, θ)

✓ + 1.5 pts Increment correct bins for each edge pixel

– 0.5 pts Iterated over edge pixels lying on each circle instead of over circles containing each edge pixel

✓ + 0.5 pts Find maximal bins

– 0.5 pts Finds all circles regardless of passing through (px, py)

+ 0 pts Incorrect

5.2

5b

1 / 2 pts

✓ + 0.5 pts $w_{ij} > w_{jk}$

✓ + 0.5 pts Intensities of i and j are more similar

+ 0.5 pts $w_{j,fg} > w_{j,bg}$

+ 0.5 pts Intensity of j is similar to foreground

+ 0 pts Incorrect or no answer

Question 6

(no title)

3.5 / 5 pts

6.1 — 6a

2.5 / 2.5 pts

✓ + 0.2 pts $u_{20} = \sum \{(x-mx)^2(y-my)^0\} / N$

✓ + 0.2 pts $u_{11} = \sum \{(x-mx)^1(y-my)^1\} / N$

✓ + 0.2 pts $u_{02} = \sum \{(x-mx)^0(y-my)^2\} / N$

✓ + 0.2 pts Invariant for translation

✓ + 0.3 pts Justify: displacement relative to mean

✓ + 0.2 pts Neither for rotation

✓ + 0.5 pts Justify: 90 deg rotation will switch x and y

✓ + 0.2 pts Covariant for scaling

✓ + 0.5 pts Justify: scaling by s, scales the moments by s^2

+ 0 pts No/Wrong answer

6.2 — 6b

0 / 1 pt

+ 0.5 pts Bins according to direction of gradient

+ 0.5 pts Increment by gradient magnitude instead of 1

✓ + 0 pts Incorrect or no answer

6.3 — 6c

1 / 1.5 pts

✓ + 0.5 pts Invariant to intensity offsets because only uses gradients

✓ + 0.5 pts Intensity scaling will scale all gradients by same amount...

+ 0.5 pts ...which will be undone by normalization

+ 0 pts Incorrect or no solution

Name: Samyak Sanghvi

Entry no: 2023CS10807

COL7(6)83 End-Semester Exam

20 November, 2025

-
- Write your name and entry number on every sheet.
 - There are 6 questions. All questions are worth 5 marks.
 - Justify all answers. Answer without justification will not receive full marks, and may receive zero.
 - All questions should be answered in the space provided. In the rare instance that you need to rewrite an answer and need more space, you can use the last page.
 - You are permitted to use one double-sided A4 sheet of handwritten notes.
-

Please sign the following declaration:

As a student of IIT Delhi, I will not give or receive aid in examinations. I will do my share and take an active part in seeing to it that others as well as myself uphold the spirit and letter of the Honour Code.

Signature: Samyak . . .

Name: Samyak Sanghvi

Entry no: 2023CS10807

1. Answer each of the following questions with a brief justification (1-2 sentences).

- (a) Starting with a low-contrast image, histogram equalization is performed, then it is performed again. What happens the second time?

No change happens as equalized image already has uniform distribution.

- (b) In colour JPEG compression, RGB values are converted to YCbCr as follows (approximately):

$$Y = 0.3R + 0.6G + 0.1B, \quad C_B = 128 + (B - Y)/1.8 \quad C_R = 128 + (R - Y)/1.4.$$

Which component(s) correspond to luminance and which to chromaticity?

$Y \rightarrow$ luminance

$C_B, C_R \rightarrow$ chromaticity.

- (c) Find the Fourier transform of a tent function $t(x) = \max(1 - |x|, 0)$, using the fact that it is the convolution of a unit box function $b(x)$ with itself.

$$b(x) = \begin{cases} 1 & -1/2 \leq x \leq 1/2 \\ 0 & \text{else} \end{cases}$$

$$b \Leftrightarrow B, \quad B = \text{sinc}(s)$$

$$t(x) = b(x) * b(x) \Leftrightarrow T(s) = (B(s))^2$$

[More space on next page...]

$$\therefore t(n) \Leftrightarrow [\text{sinc}(s)]^2$$

- (d) Consider a grayscale image corrupted by only pepper noise, i.e. some pixels are very dark. Suggest a way to eliminate the noise without affecting small/thin bright regions.

we could use the max filter / grayscale ~~dilation~~ dilation to path the dark spots

~~this~~ this won't affect the bright ^{pixels in} neighbourhoods.

- (e) When restoring a degraded image $g = h * f + \eta$, what will happen if we attempt to reduce the noise using Gaussian convolution, then apply an inverse filter to undo the blurring?

The inverse filter can't do a perfect reconstruction but approximate it

the final image would be slightly blurred but ~~not~~ less noisy.

Name: Sangyk Sanghui

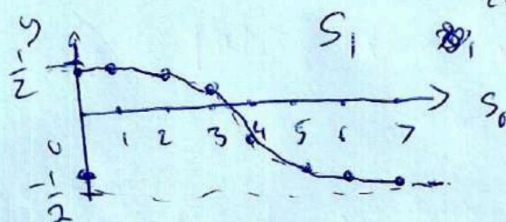
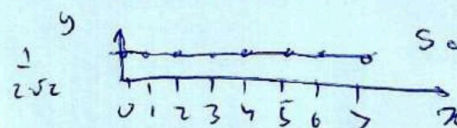
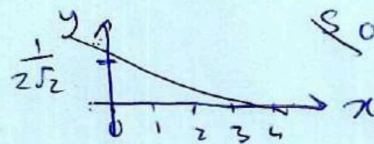
Entry no: 2024S10807

2. (a) Sketch the basis functions s_0 and s_1 for the discrete cosine transform (DCT) on a discrete signal of length 8. Label both x and y axes.

$$S_u(n) = \alpha_u \cos\left(\frac{\pi u(n+1/2)}{N}\right) \quad \alpha_u = \begin{cases} 1/\sqrt{N} & u=0 \\ \sqrt{2}/N & \text{else} \end{cases}$$

$$S_0(n) = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

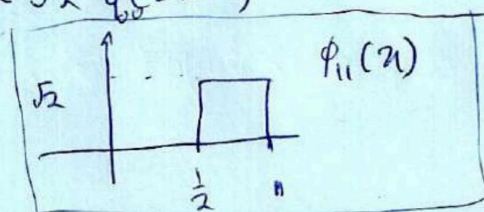
$$S_1(n) = \frac{1}{2} \cos\left(\frac{\pi(2+1)}{8}\right)$$



- (b) For the Haar wavelet transform, sketch the scaling function ϕ_{11} and the wavelet function ψ_{20} . Label both x and y axes.

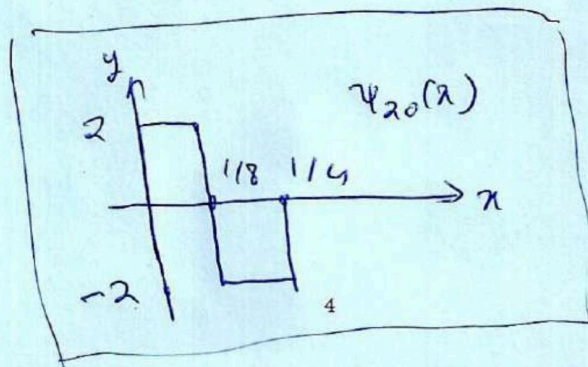
$$\phi_{00}(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 0 & \text{else} \end{cases}$$

$$\phi_{11}(x) = \frac{1}{\sqrt{2}} \phi_{00}(2x-1)$$



$$\psi_{00}(x) = \begin{cases} 1 & 0 \leq x < 1/2 \\ -1 & 1/2 \leq x < 1 \\ 0 & \text{else} \end{cases}$$

$$\begin{aligned} \psi_{20}(x) &= 2^{1/2} \psi_{10}(2^2 x - 0) \\ &= 2 \psi_{10}(4x) \end{aligned}$$



- (c) Consider (i) a smoothly varying signal with no discontinuities, (ii) a piecewise constant signal with hard boundaries between flat regions. For each of these types of signals, explain whether the DCT or the Haar wavelet transform will give a better representation, i.e. able to approximate the signal well with a small number of coefficients.

ii) The basis of Haar wavelet transform consists of small piecewise flat regions, hence by choosing the smallest non 0 flat region ^{in (i')} (let it be S) s.t. $\forall S > \frac{1}{\log_2(K+1)}$ $\{\phi_{k,j}\}_{j \in \mathbb{Z}}$ would easily with 0 error be able to ~~approx~~ obtain the function in finite coefficients.

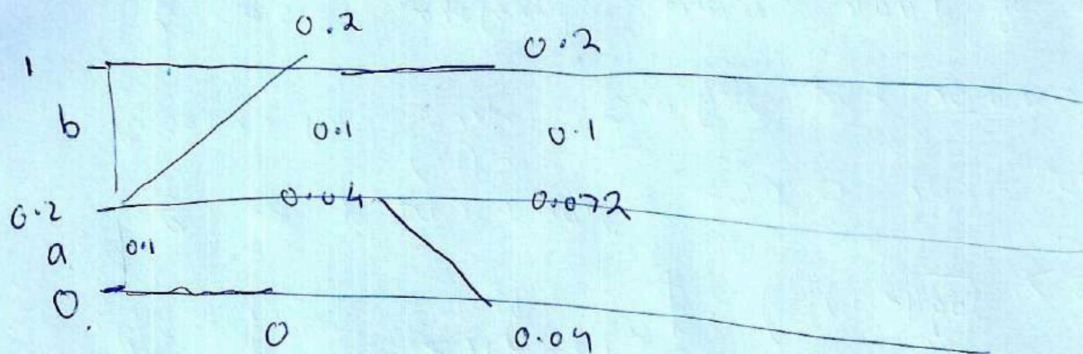
~~also for a larger resolution of Haar, few~~

i) The basis of DCT is continuous functions

This can be better used to find and obtain the approximate of the function.

for a ~~flat~~ Haar and DCT approximation with some no. of coeffs, Haar would have larger error as it grossly over/underestimates the region, while DCT would find the set of cosines best fitting the shape.

3. (a) Consider a memoryless source with two symbols a and b having probabilities $p(a) = 0.2$ and $p(b) = 0.8$. A message of length 3 was encoded using arithmetic coding, and the result was 0.1. Decode the message, showing the decoding process.



msg is abb

Since len is 3 only 3 decode steps needed.

~~step~~ step 1: $0.1 \in a$ region $\rightarrow a$

step 2: a region $0 \rightarrow 0.04$ $\Delta = 0.04$
 b region $0.04 \rightarrow 0.2$ $\Delta = 0.16$ (ratio maintained)

$0.1 \in b$ region $\rightarrow b$

step 3: a region: $0.04 \rightarrow 0.072$ $\Delta = 0.032$
 b region $0.072 \rightarrow 0.2$ $\Delta = 0.128$ (ratio maintained)
 $0.1 \in b$ region $\rightarrow b$

ans: abb.

- (b) Suppose I try to create a modified version of JPEG compression for colour images, in which I convert the image to the HSI colour space instead of YCbCr. I observe that the results are very poor, especially for red objects. Explain in detail why this occurs, describing: (i) which of the H, S, and I channels will be problematic, (ii) what will happen after block transform coding, and (iii) why the resulting coefficients will not compress well.

i) S would be problematic as it is responsible for contrast in chromaticity.

ii) During block transform coding of S, the high freq. changes across pixels would be removed.

$$S = \frac{1}{3} \min(r, g, b)$$

r+g+b

Since this is a non-linear, min function

large changes are bound to happen between pixels.

Say r is large and g, b across 2 pixels differ a little causing a significant delta in those pixels, this behaviour would be ignored.

iii) Since S is a non-linear, discontinuous metric of contrast in chromaticity this contrast is not captured well, hence essential information is lost causing bad performance.

i) cont: It can't be I as block transform works well in grayscale, features would be extracted well and it can't be h as it is a measure of blueness/greeness and slight deviation there.

4. (a) Let f be an arbitrary grayscale image, and b be the 3×3 square structuring element. Can dilation twice, $(f \oplus b) \oplus b$, be computed using a single morphological operation with a (possibly different) S.E.? Similarly, is closing twice, $(f \bullet b) \bullet b$, equivalent to a single morphological operation? Justify your answer.

1) $f \oplus b$ is same as increasing boundary by 1
 $(f \oplus b) \oplus b \Rightarrow$ increasing boundary by 2 pixels
 $\Rightarrow$ 5×5 square SE

2) $f \bullet b$ is same as patching all gaps in image of size 3×3 after doing so no such patches remain

$$\therefore (f \bullet b) \bullet b = \underline{\underline{f \bullet b}}$$

- (b) You are given a binary image A containing several non-overlapping white disks of various radii between 5 and 50 pixels. How can one use morphological operations to obtain an image containing only the disks of radius between 20 and 25 pixels?

let b_1 : disk of size 19 pixel
 b_2 : - " - 26 pixels

$(f \oplus b_1)$: ~~all~~ only discs ≥ 20 remain

$(f \oplus b_1) \ominus b_2$: only discs ≥ 26 remain

$(f \oplus b_1) - (f \oplus b_1) \ominus b_2$: ~~2 to 25~~ discs of size 20 $\rightarrow$ 25 remain

$\cap_{b_2} (f \oplus b_1)$ gives the answer.

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Name:

Samyak Singhvi

Entry no:

2023CS10807

- (c) Suppose some of the disks in the input image A are overlapping. Suggest a simple method to find the centers and radii of all the disks (regardless of size).

for i in range $[5, 50]$

temp $\leftarrow f \circ b_i$

temp1 $\leftarrow f - \text{temp}$ # has discs of size b_i only.

temp2 $\leftarrow \text{temp1} \ominus b_i$

for pixel in temp2

if pixel == 0:
continue

else:

Count[i] ++

~~pixel~~ E_i

radii[i].append(^{pixel}~~pixel~~) # the center.

$f \leftarrow f - \text{temp}$ # only has circles $> b_i$

in short using idea of (b) we obtain circles & count them by eroding them to 1 pixel. ⁹ and counting num-pixels.

5. (a) Given a binary image obtained from edge detection, design an efficient Hough transform algorithm for detecting all circles passing through a specified point (p_x, p_y) .

(If needed, you may assume a procedure is available for enumerating all pixels lying on an arbitrary line $ax + by + c = 0$.)

We maintain a 3d table with bins of

```

     $C_x, C_y, R$ 
    for pixel in image:
        pixel =  $(P_x, P_y)$ 
    for pixel in image:
        for  $C_x$  in  $1 \rightarrow m$ :
            for  $C_y$  in  $1 \rightarrow n$ :
                 $R = \sqrt{(C_x - P_x)^2 + (C_y - P_y)^2}$ 
                table
                table[ $C_x$ ][ $C_y$ ][ $R$ ] += 1
  
```

After doing this we find all ~~element~~ maximal elements in table $\geq$ threshold.

~~at~~ These elements represent the circles centered at C_x, C_y with radii R , ~~respec~~ i.e all circles in image.

~~Since we only considered 1 pixel~~ $\Rightarrow$ we then iterate over

all solⁿ C_x, C_y to see if $(P_x, P_y) \in$ circle or not. and output the ones that do.

- (b) Consider 3 adjacent pixels i, j, k in an image, with intensities $f(i) = f(j) < f(k)$, and suppose the intensity $f(j)$ is more similar to that of known foreground pixels than to background pixels. In graph cut segmentation, should the edge weight w_{ij} be greater than w_{jk} or less than it? What about the weights of edges joining pixel j with the foreground/background terminal nodes, $w_{j,fg}$ vs. $w_{j,bg}$? Justify your answer.

1) $w_{ij} = \frac{1}{\|p_i - p_j\|} \exp\left(-\frac{(I_i - I_j)^2}{2\sigma^2}\right)$ cut through i, j is not preferred.
 $w_{ij} = \frac{1}{1} e^{-0} = 1$ for min cut, it must be max.
 $w_{jk} = \frac{1}{1} e^{-d} < 1$ $\therefore \underline{w_{ij} > w_{jk}}$

2) $R_j(\text{obj}) = -\ln P(I_j | O)$

Since $f(j)$ is more similar to obj ,

$$P(I_j | O) > P(I_j | B)$$

$$\therefore \ln P(I_j | O) > \ln P(I_j | B)$$

$$\therefore -w_{j,fg} > -w_{j,bg}$$

$$\therefore \boxed{w_{j,fg} < w_{j,bg}}$$

6. (a) Give the formulas for the second-order shape moments of a region, μ_{20} , μ_{11} , μ_{02} , and discuss how they vary under translation, rotation, and scaling of the region. Are they invariant, covariant, or neither with respect to each of the three transformations?

$$\mu_{ij} = E[(x-\bar{x})^i (y-\bar{y})^j]$$

$\therefore \mu_{20} \Rightarrow$ variance along x

$\mu_{11} \Rightarrow$ covariance

$\mu_{22} \Rightarrow$ variance along y .

(1) Translation:

all 3 are invariant as ~~no~~ (since we are measuring

$$x-\bar{x}, y-\bar{y}$$

(2) Rotation.

μ_{20} , μ_{02} , μ_{11} are neither ~~invariant~~

in ~~μ_{20}~~ eg. 90° rotation swaps μ_{20} , μ_{02}

a feature centered at $0,0$ after 0

$$x, y \rightarrow x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta$$

$$\therefore E(xy) \rightarrow E(x^2 \sin \theta \cos \theta - y^2 \sin \theta \cos \theta - xy \cos^2 \theta - xy \sin^2 \theta) \neq E(xy)$$

(3) Scaling:

covariant

$$f(x) \rightarrow a f(x)$$

$$\mu_{02} \rightarrow a^2 \mu_{02} \quad \mu_{11} \rightarrow a^2 \mu_{11}$$

12

$$\mu_{20} \rightarrow a^2 \mu_{20}$$

- (b) Recall that the SIFT descriptor uses a magnitude-weighted histogram of gradient orientations in a neighbourhood of the keypoint. Explain what this means.

Say we have ~~10~~ bins of elements in range $0 \rightarrow 1$

if we get an element on the boundary of 2 bins

instead of incrementing the closest one by 1,

we increment both ~~by~~ by ratio of enclosure.

This correctly estimates in histogram how many elements lie in $a \rightarrow b$ region.

- (c) After normalization by the sum of the histogram bins, is the resulting SIFT descriptor invariant, covariant, or neither, with respect to (i) intensity offsets $f(x, y) \mapsto f(x, y) + c$, (ii) intensity scaling $f(x, y) \mapsto af(x, y)$?

i) invariant to intensity offsets of gradients are differences.

ii) Neither :

since $f(x, y) \mapsto af(x, y)$ ~~not~~ scales all gradients by 'a' too.

Hence the histogram skews more ^{to higher values} in a non linear fashion.

(Backup space for answers that need more space.)

Q3 b) i) It would be h
~~large~~ high freq. differences would be hidden / lost
 thus potentially decoding a few pixels in wrong zone.

ii) For a neighbourhood with high variance in h
 after block transform coding the large / high freq
 changes would be lost and angles approximated to be
 closer than actual.

eg. for $\theta \approx 50^\circ$; $b \approx g$ P_1 $b < g$ P_2 $b > g$

h : 50° , 310° resp but post transform
 both numbers would be somewhere in middle, potentially
 corrupting its zones.

iii) I, S are relatively continuous as large changes
 often mean finer details hence losing out due
 to compression is fine

But as we saw above example, in h for almost
 same pixels values are very far hence it
 won't compress well.